

CENG381 Stochastic Processes

Hitting Probabilities and Expected Hitting Times - Answer Key 1

Q1 Solution

a) Transition matrix P:

	1	2	3	4
St 1	[1/2	1/2	0.0	0.0]
St 2	[1/2	0.0	1/2	0.0]
St 3	[0.0	1/2	0.0	1/2]
St 4	[0.0	0.0	0.0	1.0]

b) System of equations for $\tau(i) = E[\text{steps to reach 4} \mid \text{start at } i]$:

Condition on the first step from each state:

$$\tau(1) = 1 + (1/2)\tau(1) + (1/2)\tau(2)$$

[1 step taken, then either still at 1 (prob 1/2)
or moved to 2 (prob 1/2)]

$$\tau(2) = 1 + (1/2)\tau(1) + (1/2)\tau(3)$$

[1 step taken, then moved to 1 or 3 with equal prob]

$$\tau(3) = 1 + (1/2)\tau(2) + (1/2) \cdot 0$$

[1 step taken, then moved to 2 (prob 1/2) or absorbed at 4 (prob 1/2)]

$$\tau(4) = 0 \text{ (already at target)}$$

c) Solve the system:

Rearrange each equation by moving terms with τ to the left:

$$\tau(1) - (1/2)\tau(1) = 1 + (1/2)\tau(2)$$

$$(1/2)\tau(1) = 1 + (1/2)\tau(2)$$

$$\tau(1) = 2 + \tau(2) \quad \dots (i)$$

$$\tau(3) = 1 + (1/2)\tau(2) \quad \dots (ii)$$

$$\tau(2) = 1 + (1/2)\tau(1) + (1/2)\tau(3) \quad \dots (iii)$$

Substitute (i) into (iii):

$$\tau(2) = 1 + (1/2)(2 + \tau(2)) + (1/2)\tau(3)$$

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$$\tau(2) = 1 + 1 + (1/2)\tau(2) + (1/2)\tau(3)$$

$$(1/2)\tau(2) = 2 + (1/2)\tau(3) \quad \dots \text{(iv)}$$

Substitute (ii) into (iv):

$$(1/2)\tau(2) = 2 + (1/2)(1 + (1/2)\tau(2))$$

$$(1/2)\tau(2) = 2 + 1/2 + (1/4)\tau(2)$$

$$(1/4)\tau(2) = 5/2$$

$$\tau(2) = 10$$

Back-substitute:

$$\tau(3) = 1 + (1/2)10 = 6$$

$$\tau(1) = 2 + 10 = 12$$

Answer: $\tau(2) = 10$ steps on average to reach state 4 from state 2.

Q2 Solution

a) Transition matrix P (states 0, 1, 2, 3):

	0	1	2	3
St 0	[1/2	1/2	0.0	0.0]
St 1	[0.0	1/2	1/2	0.0]
St 2	[1/2	0.0	0.0	1/2]
St 3	[0.0	0.0	0.0	1.0]

b) System of equations for $\tau(i) = E[\text{flips to reach state 3} \mid \text{start at } i]$:

$$\tau(0) = 1 + (1/2)\tau(0) + (1/2)\tau(1)$$

[flip T -> stay at 0; flip H -> go to 1]

$$\tau(1) = 1 + (1/2)\tau(1) + (1/2)\tau(2)$$

[flip H -> stay at 1; flip T -> go to 2]

$$\tau(2) = 1 + (1/2)\tau(0) + (1/2)0$$

[flip T -> back to 0; flip H -> go to 3 (done)]

$$\tau(3) = 0 \quad (\text{pattern complete, no more flips needed})$$

c) Solve for $\tau(0)$:

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Simplify each equation:

$$\begin{aligned}\tau(0): (1/2)\tau(0) &= 1 + (1/2)\tau(1) \\ \Rightarrow \tau(0) &= 2 + \tau(1) \quad \dots (i)\end{aligned}$$

$$\begin{aligned}\tau(1): (1/2)\tau(1) &= 1 + (1/2)\tau(2) \\ \Rightarrow \tau(1) &= 2 + \tau(2) \quad \dots (ii)\end{aligned}$$

$$\tau(2) = 1 + (1/2)\tau(0) \quad \dots (iii)$$

Substitute (i) and (ii) into (iii):

$$\begin{aligned}\tau(2) &= 1 + (1/2)(2 + \tau(1)) \\ &= 1 + 1 + (1/2)\tau(1) \\ &= 2 + (1/2)(2 + \tau(2)) \\ &= 2 + 1 + (1/2)\tau(2) \\ (1/2)\tau(2) &= 3 \Rightarrow \tau(2) = 6\end{aligned}$$

Back-substitute:

$$\begin{aligned}\tau(1) &= 2 + 6 = 8 \\ \tau(0) &= 2 + 8 = 10\end{aligned}$$

Answer: On average, it takes 10 coin flips to see the pattern HTH.

Q3 Solution

a) Transition probabilities:

When the collector already has i distinct coupons ($i < n$):

$$\begin{aligned}P(i, i+1) &= (n - i) / n \\ &\text{[probability that the new box contains a coupon not yet seen]}\end{aligned}$$

$$\begin{aligned}P(i, i) &= i / n \\ &\text{[probability that the new box contains a duplicate]}\end{aligned}$$

The chain has absorbing state n (all coupons collected).

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b) Geometric distribution for W_i :

When currently holding i distinct coupons, each new box independently gives a new coupon with probability $p_i = (n-i)/n$, or a duplicate with probability $1 - p_i = i/n$.

W_i = number of boxes until the next new coupon is found.

Each box is an independent Bernoulli trial with success prob p_i .

=> W_i is Geometric(p_i), and $E[W_i] = 1/p_i = n/(n-i)$.

c) Total expected time $E[T]$:

$$T = W_0 + W_1 + W_2 + \dots + W_{\{n-1\}}$$

(total boxes = sum of waiting times for each new coupon)

By linearity of expectation:

$$\begin{aligned} E[T] &= E[W_0] + E[W_1] + \dots + E[W_{\{n-1\}}] \\ &= n/n + n/(n-1) + n/(n-2) + \dots + n/1 \\ &= n * (1/n + 1/(n-1) + \dots + 1/1) \\ &= n * (1 + 1/2 + 1/3 + \dots + 1/n) \text{ (OK)} \end{aligned}$$

For $n = 5$:

$$\begin{aligned} E[T] &= 5 * (1 + 1/2 + 1/3 + 1/4 + 1/5) \\ &= 5 * (60/60 + 30/60 + 20/60 + 15/60 + 12/60) \\ &= 5 * (137/60) \\ &= 685/60 \end{aligned}$$

~ 11.42 boxes on average to collect all 5 distinct coupons.